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|--------------------------------------------------------|-------------------------------|------------------------------------------------|
| This is sample data provided for CALIBER purposes only | <b>ONE POINT LESSON (OPL)</b> | OPL No: <b>OPL-GA-1201A-04</b><br>Sheet 1 of 1 |
|--------------------------------------------------------|-------------------------------|------------------------------------------------|

|                          |                                                              |                      |                           |
|--------------------------|--------------------------------------------------------------|----------------------|---------------------------|
| <b>OPL Title</b>         | <b>Minimum Flow Line Operation &amp; Deadhead Protection</b> | <b>Discipline</b>    | Process / Operations      |
| <b>Equipment</b>         | GA-1201A - HEXANE FEED PUMP                                  | <b>Area / Unit</b>   | 1200 - HEXANE FEED SYSTEM |
| <b>Related Interlock</b> | SEQ-1201 (SIL 1)                                             | <b>P&amp;ID; Ref</b> | TJC-LLD-PID-1201          |

|                        |                                                     |                                      |                                       |                       |
|------------------------|-----------------------------------------------------|--------------------------------------|---------------------------------------|-----------------------|
| <b>Classification:</b> | <input checked="" type="checkbox"/> Basic Knowledge | <input type="checkbox"/> Improvement | <input type="checkbox"/> Trouble Case | Aspect: P-Q-C-D-S-M-E |
|------------------------|-----------------------------------------------------|--------------------------------------|---------------------------------------|-----------------------|

## 1. PURPOSE / OBJECTIVE

This One Point Lesson explains **minimum flow line operation & deadhead protection** for GA-1201A (HEXANE FEED PUMP) in the HEXANE FEED SYSTEM. Never run the pump against a closed discharge. The min-flow recycle (FV-1201) protects the pump from deadhead overheating. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

## 2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **n-Hexane** at up to 16 barg / 80 degC. Observe the following before and during the task:

- Obtain the correct operating permit and brief the shift team before the task.
- Confirm hazardous fluids are isolated / inerted as required by procedure.
- Beware of flammable, hot, or high-pressure fluids; wear the correct PPE.
- Never defeat a safety interlock or trip without an authorised override permit.
- Have emergency actions and communications ready before starting.

## 3. TOOLS & MATERIALS REQUIRED

- Operating procedure & P&ID;
- Radio / communication with panel
- Gas detector (O2 / LEL) as required
- Correct PPE for the fluid handled
- Sample containers & PPE for sampling
- Permit-to-work documentation

## 4. DETAILED PROCEDURE / STEPS

| Step | Action                                                                           | Check / Acceptance         |
|------|----------------------------------------------------------------------------------|----------------------------|
| 1    | Confirm FV-1201 min-flow recycle line is open before start                       | Permit valid & briefed     |
| 2    | Watch that FIC-1201 stays above the 9 m3/h minimum continuous flow               | Parameter within limit     |
| 3    | If FSLL-1201 trips, check downstream XV-1201 and manual valves for a closed path | Interlock status confirmed |
| 4    | Do not defeat the min-flow interlock to force higher discharge pressure          | Change logged on DCS       |
| 5    | Reset the pump only after the low-flow cause has been cleared                    | Handover to next shift     |

## 5. COMMON PROBLEMS & TROUBLESHOOTING

| Symptom                                                               | Likely Cause                                                          | Action                                                                |
|-----------------------------------------------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------|
| Hexane leak observed at GA-1201A seal gland during operation, seal dr | API Plan 11 flush orifice RO-1201 partially plugged causing seal face | Replaced JC T2100 mechanical seal cartridge, cleaned flush orifice, v |
| GA-1201A tripped on VSHH-1201 high vibration 7.4 mm/s                 | Angular misalignment 0.12 mm/100mm after foundation settlement        | Laser re-aligned pump/motor, re-shimmed motor feet, vibration back to |
| Bearing DE noisy with rising temperature TI-1201 trend                | Outer race spalling on DE bearing due to prolonged misalignment       | Renewed DE bearing 7310 BECBM, flushed housing, refilled ISO VG 68    |

## 6. KEY LEARNING POINTS

- Never run the pump against a closed discharge. The min-flow recycle (FV-1201) protects the pump from deadhead overheating.
- Respect the interlock SEQ-1201: e.g. PSLL-1201 trips at < 0.5 barg - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-1201 before working on GA-1201A.

|                             |                          |                          |                       |
|-----------------------------|--------------------------|--------------------------|-----------------------|
| Prepared by                 | Reviewed by (Supervisor) | Approved by (Manager)    | Date of Sharing       |
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